Victor González Morote

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LinkedIn | GitHub | ORCID | Google Scholar

PROFILE

Physicist and PhD candidate specializing in integrated photonics, optomechanics, and plasmonic biosensing. Hands-on experience in device design and simulation (COMSOL, Lumerical FDTD), optical characterization, instrument control and automation (MATLAB/LabVIEW/Python), and data analysis. Laboratory teaching experience supervising practical sessions, experimental protocols, and student assessment. Comfortable working cross-functionally (electronics, microfluidics, micro/nanofabrication) to deliver robust prototypes and data-driven results.

EXPERIENCE

Predoctoral Researcher — Sep 2024 – Present
University of Barcelona (IN2UB, MIND Group), Barcelona, Spain

- Designed and simulated optomechanical and plasmonic devices in COMSOL for sensing.
- Performed experimental characterization of 1D optomechanical nanocavities and miniaturized SPR platforms for gas sensing (VOC / ethanol detection).
- Project participation: ALLEGRO (PID2021-124618NB-C22, MICINN/AEI).

Research Support Technician — Sep 2022 – Aug 2024
Catalan Institute of Nanoscience and Nanotechnology (ICN2), Barcelona, Spain

- Member of NanoB2A (Prof. L. M. Lechuga). Projects: EROICA (PID2019-105132RB-I00) and POINTED (PDC2021-121325-I00).
- Built a plasmonic device for multiplexed, real-time biomarker detection; integration across electronics, photonics, microfluidics, and nanofabrication.
- Developed data acquisition and control workflows (LabVIEW/Python).

Research Initiation Grant — Jun 2021 – Aug 2021
Castelló Optics Research Group (GROC-UJI), Universitat Jaume I, Castelló, Spain

- Fiber interferometric systems with electro-optic modulators for sensing and spectroscopy.

TEACHING & ACADEMIC SERVICE

Laboratory Instructor — Academic year 2025–2026
Disseny Digital Bàsic (Basic Digital Design) — B.Eng. in Computer Engineering, Universitat de Barcelona

- 2 groups; 6 sessions/group; 2 h/session (24 h total).
- Digital design labs: VHDL + ModelSim, combinational/sequential logic, FSMs; debugging and student assessment.

Welcome Mentor — 2018 – 2021
Programa de mentoria d'acollida, Facultat de Física, Universitat de València

- Tutoring and guidance of incoming students throughout their first semester.

PAU Physics Evaluation Board Member — 2025 & 2026
Tribunals de Correccions i Vigilància, Catalonia, Spain

- Two consecutive years (2025 and 2026): invigilation and grading of the Physics exam; compliance with exam protocols.

EDUCATION

Ph.D. in Physics (in progress) 2024 – Present
Universitat de Barcelona (UB) — MIND Group (IN2UB), Dept. of Electronic and Biomedical Engineering

- Thesis: *Strategies for optimizing and combining plasmonic–polariton and cavity optomechanical sensors* (supervisors: Dr. D. Navarro-Urrios, Dr. M. Moreno-Sereno).

M.Sc. in Photonics 2022
Universitat Politècnica de Catalunya (UPC)

- Thesis: *Comparison of the optical properties of metallic versus high-dielectric 2D metasurfaces* (Advisor: A. Mihi, ICMAB-CSIC).
- Focus: Optical Engineering; Materials & Nanophotonics.

B.Sc. in Physics 2021
Universitat de València (UV)

- Thesis: *Introduction to photonic fibre-optic biosensors* (Advisors: M. Delgado Pinar, A. Díez Cremades; ICMUV).
- Focus: Photonics; Quantum Optics.

PUBLICATIONS & PRESENTATIONS

Journal articles

1. **Gonzalez-Morote, V.**; Sledzinska, M.; Gertych, A. P.; Jaffery, S. H. A.; Zdrojek, M.; Navarro-Urrios, D. *Evanescent absorption spectroscopy of transition metal dichalcogenides using guided photoluminescence in Si_3N_4 waveguides.* **Under review**, ACS Applied Optical Materials (manuscript ot-2026-00415w), 2026.
2. Tan, P.; Ferreira Curvelo, I.; Capolat Palomar, D.; **González-Morote, V.**; de Souza Matos, M. J.; Chacham, H.; Navarro-Urrios, D.; Rodríguez Viejo, J.; Esplandiu, M. J.; Sledzinska, M. *Probing the origin of distinctive low-frequency Raman features in WS_2.* Manuscript in preparation for Nano Letters, 2026.

Proceedings

1. Alonso-Tomás, D.; **González-Morote, V.**; Gomis-Bresco, J.; Capuj, N. E.; Navarro-Urrios, D. *Multimode mechanical lasing and synchronization in silicon optomechanical crystal oscillators.* Proc. SPIE 13578, Active Photonic Platforms (APP) 2025, 135780F (Sep 2025). doi:10.1117/12.3063135

Selected conference presentations

1. [**Poster**] **González-Morote, V.**; Alonso-Tomàs, D.; Capuj, N. E.; Romano-Rodríguez, A.; Navarro-Urrios, D. *Thermal optical and mechanical tuning in all-silicon and hybrid nanopillars.* META 2026, Dublin, Ireland, 14–17 Jul 2026.
2. [**Poster**] González-Morote, V.; *et al. Comparing thermal dissipation in monolithic and hybrid silicon/silicon dioxide nanopillars.* META 2025, Torremolinos, Spain, Jul 2025.

3. **[Talk]** Romano-Rodríguez, A.; *et al. Laser-based optical detection of volatile organic compounds using ZIF-8-functionalized grating-coupled SPR sensors.* IMCS 2025, Freiburg, Germany, Jun 2025.
4. **[Poster]** Moreno-Sereno, M.; *et al. SPR-grating sensor functionalised with ZIF-8 for ethanol detection.* I3S 2025, Barcelona, Spain, Nov 2025.
5. **[Poster/Oral]** Giménez-Conejo, M.; González-Morote, V.; *et al.* Abstract accepted, Eurosensors 2026, Zurich, Switzerland, 6–9 Sep 2026. [Víctor 2nd author — presentation pending]
6. **[Poster]** González-Morote, V.; *et al. Photonic sensors based on fiber devices: Narrow-bandwidth long-period gratings.* XIV International Workshop on Sensors and Molecular Recognition, Valencia, Spain, Jul 2021.

SKILLS

Programming & Data	Python (NumPy, SciPy, Pandas, Matplotlib), MATLAB, LabVIEW, Excel; C++ (basic), Mathematica
Modeling & Design	COMSOL Multiphysics (optical, thermal, mechanical); Lumerical FDTD (full simulation campaigns: grating-coupled SPR with ZIF-8, plasmonic arrays with surface lattice resonances; scripting via lumapi/Python)
Instrument Control	MATLAB and LabVIEW automation of optical benches: EXFO T500S tunable laser, Agilent 34401A multimeter, Anritsu MS2830A spectrum analyzer, tunable filter; National Instruments DAQ (USB-6009); Arduino (automated layer-by-layer deposition platform)
Fabrication	Cleanroom (RIE Oxford PlasmaPro 100, e-beam evaporation AJA ATC-8E), PVD thin films, SPR chips
Prototyping	3D printing (TinkerCAD adv., Fusion 360 int.), PDMS microfluidic flow cells
Office & Docs	Microsoft Office, Google Workspace, LaTeX, Git
Teaching	University laboratory teaching; protocol supervision; student assessment
Languages	Spanish—Native; English—B2 (Cambridge FCE); Catalan—C1 (JQCV)

CERTIFICATIONS & TRAINING

IDP-UB: "I tanmateix et mou: deleitar, commoure i persuadir amb la ciència" May 2026
Institut de Desenvolupament Professional, Universitat de Barcelona
- 2-hour accredited seminar on scientific communication; organized by the Faculty of Chemistry.

IDP-UB: Innovació i millora docent — Facultat de Física Jul 2025
Institut de Desenvolupament Professional, Universitat de Barcelona
- 5-hour accredited workshop on teaching innovation; organized by the Faculty of Physics and the Dept. of Condensed Matter Physics (UB).

STRETCHBIO Project Summer School Jul 2025
Universitat de Barcelona
- Two-day training school within the EU H2020 FET-OPEN project STRETCHBIO (GA 964808).

Microfluidic manufacturing of nanoparticles: Production considerations Mar 2023
Seminar by Prof. Yvonne Perrie, Strathclyde Institute of Pharmacy and Biomedical Sciences, Glasgow, UK — held at ICN2

- Attendance to seminar on microfluidic production of nanoparticles for pharmaceutical and biomedical applications.

Laser Safety Nov 2022
ProCareLight at ICN2 premises
- 4-hour course based on EN60825, EN207 and EN208; safety for class 3R/3B/4 lasers.

Micro- and Nanofabrication (Cleanroom) 2022–2024
ICN2 Nanofabrication Facility
- Operation of RIE (Oxford PlasmaPro 100) and e-beam evaporation (AJA ATC-8E Orion).

Data Acquisition with LabVIEW
LinkedIn Learning & self-learning
- ∼10 h coursework; application for acquisition and visualization of photodetector array signals; NI DAQ integration.

Jornada sobre Fibras Ópticas y Procesado de Señal Nov 2020
Grupo de Fibra Óptica, Universitat de València
- Attendance to a conference day on fiber optics and signal processing.

CERN Open Days Sep 2019
CERN, Geneva, Switzerland
- Attendance funded by the Universitat de València.

3D Printing & Microfluidics (self-learning)
ICN2 / self-directed
- Design and fabrication of custom enclosures for plasmonic biosensor prototypes (3D printing); PDMS-based flow-cell fabrication for biosensing.

MEMBERSHIPS & SERVICE

Founding Member, Optica Student Chapter Photonets UV 2020–2022
Optica Society (formerly OSA), Universitat de València
- Co-founded the chapter; organized and participated in scientific outreach activities.
- Attended the 1st OSA Spanish Chapters Workshop (May 2021).

BIST Mentoring Programme 2022 – 2024
Barcelona Institute of Science and Technology
- Participant in the inter-institutional mentoring programme.